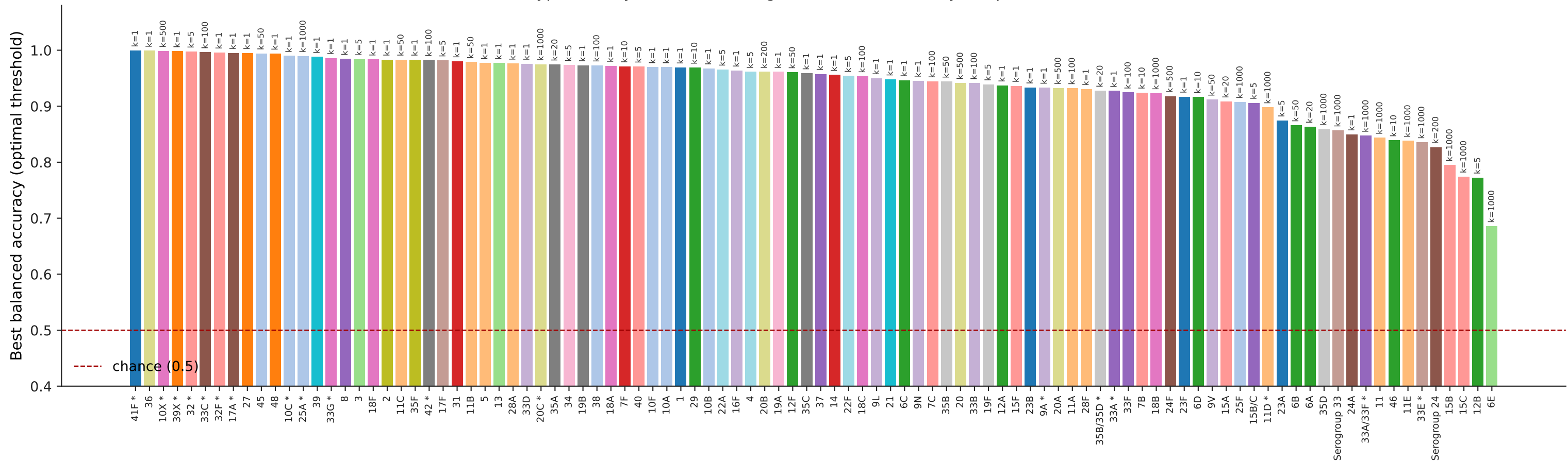


Per-serotype novelty-detection ceiling — balanced accuracy at optimal k & threshold



Ceiling: k and threshold both tuned per serotype on the held-out novels (optimistic, NOT deployable). Balanced accuracy = (TPR+TNR)/2; 0.5 = chance. * n_{novel}<5 (unreliable).